12 Ch 12 – Interest Rate Risk Management

Asked in last 3 papers: FRA settlement (Sep 2025, 4m) · cap for borrower and floor for depositor with premium (Jan 2026, 6m) · zero-cost collar (May 2026, 4m) · re-pricing gap and Earnings at Risk (May 2026, 7m).

1 Introduction

→ Interest = **cost of borrowing** and compensation for the service and **risk of lending** (default risk + **inflation** risk; also covers the lender's processing cost).

→ Firms with low margins and high capital expenses are very sensitive to rate rises; interest rate **derivatives** mitigate risk and build value.

→ **Factors affecting interest rates** (macro-economic): (a) **demand/supply of money** — high growth → higher demand → rates up; (b) **inflation** — higher inflation → rates rise; (c) **Government** — biggest borrower; RBI changes key rates (CRR, SLR, bank rate) by printing notes or Open Market Operations (OMO).

2 Benchmark Rates

→ Benchmark (**reference**) rate = rate that forms the basis for other rates: loans, overdrafts, mortgages, floating-rate loans, and derivatives (forwards, futures, options, **swaps**). Basis points (BPs) are added over it depending on credit rating. Set by an **independent body**.

→ **LIBOR** manipulated (2012) → decided in 2017 to cease; from **1 Jan 2022** contracts use **Alternative Reference Rates (ARRs)**.

→ ARR vs LIBOR: (i) ARRs are based on **actual overnight transactions** (secured or unsecured); LIBOR is unsecured and relies on **judgment of panel banks**; (ii) ARRs are **near risk-free** with **no term premium**. Each currency has its own ARR, regulator and nature.

Region	ARR	Regulator	Nature
USA	SOFR (Secured Overnight Financing Rate)	Federal Reserve Bank of New York	Secured
UK	SONIA (Sterling Overnight Index Average)	Bank of England	Unsecured
Europe	€STER (Euro Short-Term Rate)	European Central Bank	Unsecured
Japan	TONAR (Tokyo Overnight Average Rate)	Bank of Japan	Unsecured
Switzerland	SARON (Swiss Average Rate Overnight)	SIX (Swiss Stock Exchange)	Secured

→ **India:** Repo, PLR, MCLR exist, but commonest are **MIBOR** (rate bank **charges** a borrower) and **MIBID** (rate at which a bank would like to **borrow** from another bank). Used in IRS, FRA, floating rate debentures and for valuing bonds/debentures.

3 / 3.1 Interest Rate Risk; Types of Interest Rate Risk

→ Change in **prices of bonds** due to change in interest rates, plus impact on **interest income** → both **price risk** and **reinvestment risk** need focus. Banks gain or lose with every change in rates on fixed-term deposits.

3.1.1 Gap Exposure

Asked: May 2026 Q5(b) 7m

→ Gap (mismatch) risk: assets, liabilities and off-balance sheet items with different **principal amounts, maturity dates or re-pricing dates**. Most important for banks.

Formula	Terms / logic	Effect on NII
Gap = RSA − RSL	RSA = interest rate sensitive assets; RSL = interest rate sensitive liabilities (re-priced in the time band). Gap = measure of rate sensitivity.	**Positive** (asset sensitive): rate **rise** → NII up. **Negative** (liability sensitive, RSL > RSA): rate **fall** → NII up; rise → NII down.
EaR = Gap × assumed change in interest rate	Earnings at Risk: how much earnings may be hit by an adverse rate move. Rate change estimated from past trends / forecasts.	Bank fixes an EaR limit (based on last/current year income) and a **trigger point** at which on- or off-balance sheet **hedging** starts.

Gap figures can be augmented with the average coupon in each time band to estimate NII; periodic gap analysis shows exposure over distinct maturities and the portfolio change needed to alter the risk profile. (In May 2026 Q5(b) the gap was for a quarter, so EaR used × 1/4, compared with a limit of 8% of NII.)

Limitations of Gap report:

- → Measures only the **time difference** in re-pricing; ignores **basis** and **embedded option** risks.
- → Assumes all items in a time band mature/re-price **simultaneously**; ignores effect on **economic/market value** of assets, liabilities and off-balance sheet positions.
- → Ignores differences in **timing of payments** due to changed rate environment.
- → Assumes **parallel shift** in yield curve, which seldom happens.
- → Ignores variability in **non-interest revenue and expenses**.

3.1.2–3.1.6 Basis, Embedded Option, Yield Curve, Price, Reinvestment Risk

Risk	SM point
3.1.2 Basis risk	Rates on different assets, liabilities and off-balance sheet items change by **different magnitudes** (e.g. assets on fixed rate, liabilities on floating). High where composite assets are funded from composite liabilities (Indian loan book). NII expands → favourable basis shift; contracts → basis moved against the bank.
3.1.3 Embedded option risk	Big rate changes encourage **prepayment** of loans, exercise of **call/put on bonds**, **premature withdrawal** of deposits. Faster and bigger the change, greater the risk. Banks should estimate embedded options, adjust Gap statements and stipulate **penalties** based on opportunity cost.
3.1.4 Yield curve risk	Yield curve moves frequently through **business cycles**; evaluate its impact on portfolio value and income.
3.1.5 Price risk	When assets are **sold before maturity**; bond prices and yields are inversely related. Linked to the **trading book** → limits on portfolio size, holding period, duration, defeasance period, stop loss, marking to market.
3.1.6 Reinvestment risk	Uncertainty about the rate at which **future cash flows** can be reinvested; cash-flow mismatches → variation in NII.

3.2 Measuring Interest Rate Risk

- → Risk must be **identified and quantified** before it can be managed or hedged.
- → The IRR system should cover all sources (gap, basis, embedded option, yield curve, price, reinvestment, net interest position), the characteristics of each position, and the full range of rate movements.
- → **Techniques: Maturity Gap Analysis** (sensitivity of **earnings**), **Duration** (sensitivity of **capital**), **Simulation**, **Value at Risk**; often used in combination / hybrid.

4 Hedging Interest Rate Risk

(A) Traditional	(B) Modern
i. Asset and Liability Management (ALM); ii. Forward Rate Agreement (FRA)	i. Interest Rate Futures (IRF); ii. Interest Rate Options (IRO); iii. Interest Rate Swaps

4.1 / 4.1.1 Traditional Methods — Asset and Liability Management (ALM)

→ Tool evolved by **RBI**; introduced in Indian banks w.e.f. **1 April 1999**.

→ **Comprehensive and dynamic framework** for measuring, monitoring and managing market risk: managing the **structure of the balance sheet** so that **net interest earnings are maximised** within the overall risk preference (present and future).

→ Covers liquidity, interest rate, foreign exchange, equity and commodity price risks; integrated with business strategy. Functions extend to liquidity risk, market risk, trading risk, funding & capital planning, profit planning, growth projection.

→ Relevant for banks, finance, leasing and insurance companies. Gives a true picture of the **risk/reward trade-off**. A **first step in long-term strategic planning** → planning function for an **intermediate term** (planning, direction and control of levels, changes and mixes of assets, liabilities and capital).

4.1.2 Forward Rate Agreements (FRAs)

Asked: Sep 2025 Q2(b) 4m

Features (TYK T1 — short note):

→ Agreement by which a **borrower/lender** protects itself from unfavourable rate changes; **OTC**, not exchange-traded (unlike futures).

→ Used by banks to fix interest cost on anticipated deposits or revenue on variable-rate loans indexed to a benchmark (SOFR, €STER, MIBOR).

→ **Off-balance sheet**; **no transfer of principal** — principal is **notional**.

→ **Cash-settled** at maturity for the profit/loss. **Seller** pays buyer the increased interest cost if RR > FR; **buyer** pays seller if RR < FR. So a **borrower buys** an FRA (protection against a rise).

→ Differential is **discounted at the actual (post-change) rate** because it is settled at the **beginning** of the period, not at the end.

Formula	Terms / logic	SM example figures
Settlement = N × (RR − FR) × (dtm/DY) ÷ [1 + RR × (dtm/DY)]	**N** = notional principal. **RR** = reference rate for the contract maturity on the settlement date (SOFR/€STER/MIBOR). **FR** = agreed forward rate. **dtm** = FRA period in days. **DY** = day-count basis, 360 or 365. Numerator = extra (or saved) interest for the FRA period; denominator = discount at RR for the same period (paid upfront). Positive → seller pays buyer; negative → buyer pays seller.	$1 m, FR 5%, 3-month SOFR 6%: 1 bp on $1 m for 90 days = **$25** → $25 × 100 bp = **$2,500** → buyer receives $2,500 ÷ [1 + 0.06(90/360)] = **$2,463.05**.
a × b FRA	Settlement after **a** months; deposit/loan matures **b** months from contract date → FRA period = b − a months. Use RR for that (b − a) period.	**1 × 4**: deposit placed after 1 month, maturing 4 months from contract (3-month period).

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TYK P1 (Parker & Co.): borrow ₹ 60 crore for 3 months in 6 months; FRA 9.30%. (i) Actual 9.60%: 60 cr × (0.096 − 0.093)(3/12) ÷ [1 + 0.096(3/12)] = 60 cr × 0.00075 ÷ 1.024 = **₹ 4,39,453.13** → banker pays Parker. (ii) Actual 8.80%: 60 cr × (−0.00125) ÷ 1.022 = **− ₹ 7,33,855.19** → Parker pays banker. ICAI: 90/360, 90/365, 91/360 or 91/365 also accepted.

TYK P2 (TM Fincorp, bought 6 × 9, ₹ 100 crore at 5.25%): 6 × 9 = 3-month rate in 6 months → use 3-month MIBOR **5.50%** (not 6 or 9-month). (a) Profit **25 basis points**. (b) 100 cr × (5.50% − 5.25%)(90/360) ÷ [1 + 0.055(90/360)] = **₹ 6,16,523** profit.

4.2 / 4.2.1 Modern Methods — Interest Rate Futures (IRF)

→ Futures contract on an **interest-bearing asset** (future delivery); locks in the price of that asset for a future date. **Bonds** are the underlying, not the interest rate.

→ Logic: rates and bond prices move **inversely**. Borrower fears a rise → **sells futures** today; if rates rise, futures price falls → profit on closing out (buying back).

→ NSE IRF: futures on **6, 10 and 13 year GOI Security** and **91-day T-Bill (91DTB)**. Settlement at two levels: **daily MTM** and **physical delivery** on any day in the expiry month; final settlement only on expiry date. Prices by demand and supply.

Formula	Terms / logic
Delivery price = Conversion factor × Futures price	**Conversion factor** (introduced by RBI, published by NSE per bond per expiry) makes deliverable bonds of different maturities/coupons comparable with each other and with the notional bond.
Seller's profit = (Futures settlement price × CF) − Quoted spot price; Seller's loss = Quoted spot price − (Futures settlement price × CF)	**Cheapest to Deliver (CTD)** = bond that minimises the gap between quoted spot price and CF-adjusted futures settlement price, i.e. the bond that **maximises profit or minimises loss** of the futures seller — least expensive in the deliverable basket.

4.2.2 Interest Rate Options (IRO)

Asked: Jan 2026 Q4(a) 6m

→ Also called **Interest Rate Guarantee (IRG)**; a **right, not an obligation**; acts as **insurance** — protection against adverse moves while keeping benefit of favourable moves. Cost = **premium**.

→ An IRO is basically a **series of FRAs** exercisable at pre-set benchmarked rates for each period (3 months, 6 months ...).

4.2.2.1–4.2.2.2 Cap Option and Floor Option

Asked: Jan 2026 Q4(a) 6m

→ **Cap** (call option): buyer pays premium for the right to receive the excess interest whenever the index rises **above the cap rate**; no obligation if rates fall. Bought by a **borrower** on floating-rate debt. Evolved from guarantees on floating-rate loans; OTC caps from **1985** (banks stripped guarantees from FRNs); spurred by the **leveraged buyout** boom of the 1980s (lenders required LBO borrowers to buy caps); activity declined as LBOs fell.

→ **Cap is characterised by: notional principal**; **benchmark** (SOFR, MIBOR, PLR) of a specified maturity; **cap rate** (= strike); **period** incl. payment and reset dates. Payments follow swap conventions; reset intervals usually equal the benchmark's term.

→ **First period is excluded** — its rate is known on the agreement date.

→ **Floor** (OTC): protects the buyer against a **fall** in rates — bought by a **lender/depositor**. Payoff when rate < floor rate = payoff from **selling an FRA**; above floor → nothing, premium lost. Floor = **European put on an FRA**; multi-period floor = bundle of European puts on a sequence of FRAs.

Formula	Terms / logic	SM example figures
Cap payoff = N × max(0, $r - r_c$) × (dt / days in year)	r = actual spot rate on reset date; r_c = cap rate (decimal); **dt** = days from reset date to payment date. Paid on the **next payment date** (in arrears).	$1 m, 6-month SOFR cap 5%, 15 Jan–15 Jan, reset 15 July; SOFR 5.5% → on next 15 Jan (184 days): $1,000,000 × (0.055 − 0.050)(184/360) = **$2,555.56**.
Floor payoff = N × max(0, $r_f - r$) × (dt / days in year)	r_f = floor (strike) rate; dt = days from last reset to payment date.	—
Premium per reset period = (Lump-sum premium rate × A) ÷ PVAF(i, t), i.e. $P = rp \times A \div [(1/i) \times (1 - 1/(1+i)^t)]$	Upfront lump-sum premium spread into equal amounts per reset period. **A** = principal; **rp** = premium rate; **i** = fixed rate per period; **t** = number of periods.	TYK P3 below.

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TYK P3 (XYZ Ltd): £ 15 m, 6-month SONIA, 24 months; cap 8%; lump sum premium 1% (= £ 150,000); fixed rate 7% p.a. → 3.5% per half-year, 4 periods. PVAF(3.5%, 4) = 0.966 + 0.933 + 0.901 + 0.871 = **3.671** → premium **£ 40,861** per period (formula route: **£ 40,595**). SONIA 9%, 9.5%, 10% → amount from bank = (SONIA − 8%) × 15 m × 6/12 = **£ 75,000; £ 112,500; £ 150,000 = £ 337,500**. Less premium 3 × 40,861 = £ 122,583 → net **£ 214,917** (with £ 40,595: **£ 215,715**). Risk of £ 337,500 reduced by £ 214,917.

TYK P7 (textile manufacturer): floating loan ₹ 40,00,000; to peg cost at 8.5%, **buy caps**: notional ₹ 40,00,000, strike 8.5%, settlement 31 March each year till 31 March 2016. Premium 0.75% paid once on 1 Oct 2012 = **₹ 30,000**. Compensation: 2013 (10.2%) 1.70% → **₹ 68,000**; 2014 (11.5%) 3.00% → **₹ 1,20,000**; 2015 (9.25%) 0.75% → **₹ 30,000**; 2016 (8.25% < 8.5%) → **nil**. Cost never exceeds 8.5%.

Note (app-written): TYK P3 states the loan as "SONIA + 10.00%"; the spread does not affect the cap payoff (only SONIA vs 8% matters). ICAI spreads the premium over 4 half-years (PVAF 3.5%, 4) but deducts it for only the 3 reset periods shown (first period excluded), so the premium deducted totals only £ 122,583 (three of four instalments). The exact PVAF(3.5%, 4) is 3.6731, giving about £ 40,838; ICAI's £ 40,861 uses the rounded 3.671 and its alternative £ 40,595 does not reconcile exactly. Keep ICAI's figures.

4.2.2.3 Interest Rate Collars

Asked: May 2026 Q4(c) 4m

- Collar = **buy a Cap + sell a Floor** → floating rate **bounded** on both high and low side.
- Limits benefit from a big fall in rates, but **cheaper than a cap alone**: floor premium received offsets cap premium.
- **Zero-cost collar:** premium earned on the floor **exactly offsets** the cap premium (value of cap = value of floor; solve for the unknown floor or cap rate).

Formula	Terms / logic
Collar payment = N × [max(0, r − r_c) − max(0, r_f − r)] × (dt / days in year)	First bracket = cap receipt (r above cap); second = floor payment owed (r below floor); between r_f and r_c → nil. r = actual spot rate on reset date; dt = term of the index in days.

4.2.3 Interest Rate Swaps; 4.2.3.1–4.2.3.4 Dealers, characteristics, timing, price quotation

- **Swap counterparties** exchange payments indexed to **two different interest rates** on a **notional principal** that is **never exchanged**.
- **4.2.3.1 Swap dealers:** intermediary took a **brokerage fee**, no continuing role; contract between the two ultimate users, who paid each other directly.
- **4.2.3.2 Fixed/floating swap is characterised by:** a **fixed rate**; a **floating rate** periodically reset; a **notional principal**; the **term**, with schedule of **reset dates** and **payment dates**.
- **4.2.3.3 Timing:** negotiated on **trade date**; takes effect **two days later** = **settlement date**.
- **4.2.3.4 Price quotation (swap rate):** fixed rate quoted as **All-In-Cost (AIC)** — the fixed rate relative to a **flat** floating index (e.g. "8% against six month SOFR flat").

4.2.3.5 Types of Swap

Type	SM point
(a) Plain vanilla (generic / coupon swap)	Exchange of **fixed** for **floating** over a period on notional principal; floating basis SOFR, €STER, SONIA, MIBOR, PLR. Fixed leg: 30-day months, 360-day year.
(b) Basis rate swap (non-generic)	**Both legs floating**, on different benchmarks (e.g. 1-month SOFR vs 3-month SOFR).
(c) Asset swap	Exchange of **fixed-rate investments** (bonds with guaranteed coupon) for **floating-rate investments** (an index).
(d) Amortising swap	Notional principal **declines** over the life; useful for issuers of **redeemable bonds/debentures** — hedges rate risk in line with the redemption profile.

Formula (plain vanilla)	Terms / logic	SM example figures
Semi-annual fixed payment = N × AIC × (180/360)	N = notional; AIC = quoted fixed rate (swap rate). 30/360 basis → always 180/360 per half-year.	**TYK P4:** AIC 8% vs 6-month SOFR flat, US$ 5,00,000. (i) Fixed = 5,00,000 × 0.08 × 0.5 = **20,000**. (ii) Floating, 181 days, SOFR 6% = 5,00,000 × 0.06 × 181/360 = **US$ 15,090** (using 0.503) or **15,083** (0.502777). Net: fixed payer pays **US$ 4,910** (or 4,917).
Floating payment = N × R × (dt/360)	R = reference rate (SOFR, €STER, SONIA, MIBOR) set at the start of the period; dt = **actual days** since last settlement.	see TYK P4
Net settlement = Fixed leg − Floating leg	Only the difference is paid, by the party owing the larger leg.	see TYK P4

Note (app-written): ICAI's TYK P4 answer prints the fixed payment as "₹ 20,000/-"; the swap is in US$, so read it as US$ 20,000.

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TYK P5 (Derivative Bank, OIS): receive MIBOR overnight (compounded daily on 365 days), pay fixed, ₹ 10 crore, 7 days from Tue 3 Aug 2010; Sunday holiday (Saturday rate 7.98% applied for 2 days). Daily interest on growing principal: 21,233; 22,334; 22,256; 21,795; 43,764 (2 days); 22,358 → floating total **₹ 1,53,740**. Less net received ₹ 317 → fixed leg **₹ 1,53,423** → fixed rate = 1,53,423 ÷ 10 crore × 365/7 = 0.07999914 ≈ **8%**. (360-day basis also accepted.)

The SM chapter gives no separate formula for valuing an existing swap; TYK P5 values both legs and solves the fixed rate from the net settlement.

Comparative advantage (TYK P6, a currency swap placed in this chapter):

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TYK P6 (A Inc / B Inc): each needs $ 200,000 (B in ¥) for 1 year; rate $1 = ¥120. Rates: A — ¥ 5%, $ 9%; B — ¥ 8%, $ 10%. A borrows ¥, B borrows $, swap; B pays A 6% (1% over A's ¥ rate 5%), A pays B only 9%.

A Inc: pays ¥ lender ¥ 12,00,000; receives from B ¥ 14,40,000; pays B $ 18,000 (¥ 21,60,000) → net ¥ 19,20,000 = **$ 16,000 (8%)** vs $ 18,000 without swap → gain **$ 2,000 (1%)**.

B Inc: pays $ lender $ 20,000; receives from A $ 18,000; pays A ¥ 14,40,000 ($ 12,000) → net $ 14,000 = **¥ 16,80,000 (7%)** vs ¥ 19,20,000 (8%) without swap → gain **¥ 2,40,000 (1%)**.

Logic (app-written, from TYK P6 figures): A is cheaper in both currencies but by 3% in ¥ and only 1% in $; each borrows where its relative advantage lies (A in ¥, B in $) and the 2% difference is shared, 1% each.

4.2.4 Swaptions; 4.2.4.1–4.2.4.3 Features, pricing, uses

→ Option on an interest rate swap: right, not obligation, to enter an IRS at a **specific future date**, at a **particular fixed rate**, for a **specified term**. (TYK T2)

→ **Fixed rate payer swaption (Call swaption):** right to **pay fixed, receive floating**. **Fixed rate receiver swaption (Put swaption):** right to **receive fixed, pay floating**.

4.2.4.1 Principal features:

→ A. Option on a **forward-start IRS**; fixed rate, floating reference rate and tenure fixed when the swaption is concluded.

→ B. **3-month into 5-year** swaption = option to enter a 5-year IRS, 3 months from now.

→ C. **Option period** = time from transaction date to expiry date.

→ D. Premium expressed in **basis points**.

→ E. Can be **cash-settled**: at expiry marked to market off the applicable **forward curve**, difference settled in cash.

4.2.4.2 Pricing: model of the **probability distribution of the forward zero-coupon curve** which undoes a market process.

4.2.4.3 Uses (SM prints this as "2.2.4.3"):

→ Applied by both **active traders** and **corporate treasurers**.

→ Swap traders: **speculation** or to **hedge part of swap books**.

→ Hedging **embedded optionality** in many businesses.

→ Borrowers targeting an **acceptable borrowing rate**.

→ Businesses **tendering for contracts**.

→ Protection on **callable/puttable bond issues**.

Exam steps — FRA: identify buyer (borrower) / seller → FRA period from a × b (b − a months) and pick RR for that period → (RR − FR) × N × dtm/DY → ÷ [1 + RR × dtm/DY] → sign: + seller pays buyer, − buyer pays seller → state who pays whom.

Exam steps — Cap / Floor: borrower buys cap, depositor buys floor → premium = rp × N, spread per period ÷ PVAF(fixed rate per period, n) → each reset: payoff = N × max(0, $r - r_c$) [or max(0, $r_f - r$)] × period → net = payoff − premium per period → total net gain / risk reduced.

Exam steps — Collar: buy cap + sell floor → payoff = cap receipt − floor payment each period → zero-cost: PV of cap = PV of floor → solve unknown rate → effective rate lies between floor and cap.

Exam steps — Plain vanilla / OIS: fixed leg N × AIC × 180/360 → floating leg N × R × actual days/360 (OIS: compound daily, holidays carry previous rate) → net = difference → solve fixed rate from net if given.

Exam steps — Comparative advantage swap: table of each party's rates → each borrows where relatively cheaper → list receipts/payments under swap → net cost vs cost without swap → gain to each.

Exam steps — Gap / EaR: RSA − RSL per time band → sign (positive gains on rise, negative gains on fall) → EaR = gap × rate change × fraction of year → compare with EaR limit → hedge if beyond trigger.

Past papers

⑫ Interest Rate Risk Management — what ICAI asked (Sep 2025, Jan 2026, May 2026)

21 marks *set across the last three attempts (5.9% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	Q2(b)	4	FRA final settlement amount for a borrower, for a higher and a lower actual rate (formula with days/year basis) (Numerical)
Jan 2026	Q4(a)	6	Interest rate options: strategy for borrower (buy cap) and depositor (buy floor), premium from lump-sum premium rate, net gain/loss from hedging (Numerical)
May 2026	Q4(c) [or]	4	Zero-cost collar on a floating-rate loan: value of cap = value of floor (intrinsic value, PV of payoffs); solve the floor rate (Numerical)
May 2026	Q5(b)	7	Re-pricing gap (RSA - RSL): negative gap, favourable/unfavourable scenario, Earnings at Risk = gap x rate change x 1/4 versus 8% of NII limit, hedging decision (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Write a short note on Forward Rate Agreements.

ICAI answer: "Please refer paragraph 4.1.2" – SM paragraph 4.1.2 reproduced

4.1.2 Forward Rate Agreements (FRAs)

A Forward Rate Agreement (FRA) is an agreement between two parties through which a borrower/ lender protects itself from the unfavourable changes to the interest rate. Unlike futures FRAs are not traded on an exchange thus are called OTC product. Following are main features of FRA.

- Normally it is used by banks to fix interest costs on anticipated future deposits or interest revenues on variable-rate loans indexed to Benchmark Interest Rate e.g. SOFR, ESTER, MIBOR etc.
- It is an off-Balance Sheet instrument.
- It does not involve any transfer of principal. The principal amount of the agreement is termed "notional" because, while it determines the amount of the payment, actual exchange of the

principal never takes place.

- It is settled at maturity in cash representing the profit or loss. A bank that sells an FRA agrees to pay the buyer the increased interest cost on some "notional" principal amount if Reference Rate of some specified maturity is above a stipulated "Forward Interest Rate" on the contract maturity or settlement date. Conversely, the buyer agrees to pay the seller any decrease in interest cost if Reference Rate fall below the forward rate.
- Final settlement of the amounts owed by the parties to an FRA is determined by the formula

$$= \frac{(N)(RR - FR)(dtm/DY)}{[1 + RR(dtm/DY)]} \times 100$$

Where,

N = the notional principal amount of the agreement;

RR = Reference Rate for the maturity specified by the contract prevailing on the contract settlement date; typically SOFR, ESTER or MIBOR

FR = Agreed-upon Forward Rate; and

dtm = maturity of the forward rate, specified in days (FRA Days)

DY = Day count basis applicable to money market transactions which could be 360 or 365 days.

If Reference Rate > FR then the seller owes the payment to the buyer, and if Reference Rate < FR then the buyer owes the seller the absolute value of the payment amount determined by the above formula.

- The differential amount is discounted at post change (actual) interest rate as it is settled in the beginning of the period not at the end.

Example

Suppose two banks enter into an agreement specifying:

- a forward rate of 5 percent on a Eurodollar deposit with a three-month maturity;
- a $1 million notional principal; and settlement in one month.

Such an agreement is termed a 1x4 FRA because it fixes the interest rate for a deposit to be placed after one month and maturing four months after the date the contract is negotiated.

If the three-month SOFR is 6 percent on the contract settlement date, the seller would owe the buyer the difference between 6 and 5 percent interest on $1 million for a period of 90 days.

Every 1 basis point change in the interest rate payable on a principal of $1 million for a 90-day maturity changes interest cost by $25, so that the increase in the interest cost on a three-month Eurodollar deposit over the specified forward rate in this case is $25 x 100 basis points = $2,500.

The $2,500 difference in interest costs calculated above is discounted back three months using the actual three-month SOFR prevailing on the settlement date.

Thus, if 90-day SOFR turns out to be 6 percent on the contract maturity date the buyer would receive $2,463.05 = $2,500/[1 + 0.06(90/360)].

ICAI answer: "Please refer paragraph 4.1.2" – SM paragraph 4.1.2 reproduced

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- It is an off-Balance Sheet instrument.
- It does not involve any transfer of principal. The principal amount of the agreement is termed "notional" because, while it determines the amount of the payment, actual exchange of the principal never takes place.
- It is settled at maturity in cash representing the profit or loss. A bank that sells an FRA agrees to pay the buyer the increased interest cost on some "notional" principal amount if Reference Rate of some specified maturity is above a stipulated "Forward Interest Rate" on the contract maturity or settlement date. Conversely, the buyer agrees to pay the seller any decrease in interest cost if Reference Rate fall below the forward rate.
- Final settlement of the amounts owed by the parties to an FRA is determined by the formula

$$= \frac{(N)(RR - FR)(dtm/DY)}{[1 + RR(dtm/DY)]} \times 100$$

Where,

N = the notional principal amount of the agreement;

RR = Reference Rate for the maturity specified by the contract prevailing on the contract settlement date; typically SOFR, ESTER or MIBOR

FR = Agreed-upon Forward Rate; and

dtm = maturity of the forward rate, specified in days (FRA Days)

DY = Day count basis applicable to money market transactions which could be 360 or 365 days.

If Reference Rate > FR then the seller owes the payment to the buyer, and if Reference Rate < FR then the buyer owes the seller the absolute value of the payment amount determined by the above formula.

- The differential amount is discounted at post change (actual) interest rate as it is settled in the beginning of the period not at the end.

Example

Suppose two banks enter into an agreement specifying:

- a forward rate of 5 percent on a Eurodollar deposit with a three-month maturity;
- a $1 million notional principal; and settlement in one month.

Such an agreement is termed a 1x4 FRA because it fixes the interest rate for a deposit to be placed after one month and maturing four months after the date the contract is negotiated.

If the three-month SOFR is 6 percent on the contract settlement date, the seller would owe the buyer the difference between 6 and 5 percent interest on $1 million for a period of 90 days.

Every 1 basis point change in the interest rate payable on a principal of $1 million for a 90-day maturity changes interest cost by $25, so that the increase in the interest cost on a three-month Eurodollar deposit over the specified forward rate in this case is $25 x 100 basis points = $2,500.

The $2,500 difference in interest costs calculated above is discounted back three months using the actual three-month SOFR prevailing on the settlement date.

Thus, if 90-day SOFR turns out to be 6 percent on the contract maturity date the buyer would receive $2,463.05 = $2,500/[1 + 0.06(90/360)].

Theory Q2

2. What do you know about swaptions and their uses?

ICAI answer: "Please refer paragraph 4.2.4" – SM paragraph 4.2.4 reproduced

4.2.4 Swaptions

An interest rate swaption is simply an option on an interest rate swap. It gives the holder the right but not the obligation to enter into an interest rate swap at a specific date in the future, at a particular fixed rate and for a specified term.

There are two types of swaption contracts: -

- A **fixed rate payer swaption** (also called Call Swaption) gives the owner of the swaption the right but not the obligation to enter into a swap where they pay the fixed leg and receive the floating leg.
- A **fixed rate receiver swaption** (also called Put Swaption) gives the owner of the swaption the right but not the obligation to enter into a swap in which they will receive the fixed leg, and pay the floating leg.

4.2.4.1 Principal features of Swaptions

A. A swaption is effectively an option on a forward-start IRS, where exact terms such as the fixed rate of interest, the floating reference interest rate and the tenure of the IRS are established upon conclusion of the swaption contract.

B. A 3-month into 5-year swaption would therefore be seen as an option to enter into a 5-year IRS, 3 months from now.

C. The 'option period' refers to the time which elapses between the transaction date and the expiry date.

D. The swaption premium is expressed as basis points.

E. Swaptions can be cash-settled; therefore, at expiry they are marked to market off the applicable forward curve at that time and the difference is settled in cash.

4.2.4.2 Pricing of Swaptions

The pricing methodology depends upon setting up a model of probability distribution of the forward zero-coupon curve which undoes a Market process.

2.2.4.3 Uses of Swaptions

(a) Swaptions can be applied in a variety of ways for both active traders as well as for corporate treasurers.

(b) Swap traders can use them for speculation purposes or to hedge a portion of their swap books.

(c) Swaptions have become useful tools for hedging embedded optionality which is common to the natural course of many businesses.

(d) Swaptions are useful to borrowers targeting an acceptable borrowing rate.

(e) Swaptions are also useful to those businesses tendering for contracts.

(f) Swaptions also provide protection on callable/puttable bond issues.

ICAI answer: "Please refer paragraph 4.2.4" – SM paragraph 4.2.4 reproduced

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Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical QI

1. M/s. Parker & Co. is contemplating to borrow an amount of ₹60 crores for a Period of 3 months in the coming 6 month's time from now. The current rate of interest is 9% p.a., but it may go up in 6 month's time. The company wants to hedge itself against the likely increase in interest rate.

   The Company's Bankers quoted an FRA (Forward Rate Agreement) at 9.30% p.a.

   What will be the Final settlement amount, if the actual rate of interest after 6 months happens to be (i) 9.60% p.a. and (ii) 8.80% p.a.?

Solution — ICAI

1. Final settlement amount shall be computed by using formula:

$$= \frac{(N)(RR - FR)(dtm/DY)}{[1 + RR(dtm/DY)]}$$

Where,

N = the notional principal amount of the agreement;

RR = Reference Rate for the maturity specified by the contract prevailing on the contract settlement date;

FR = Agreed-upon Forward Rate; and

dtm = maturity of the forward rate, specified in days (FRA Days)

DY = Day count basis applicable to money market transactions which could be 360 or 365 days.

Accordingly,

If actual rate of interest after 6 months happens to be 9.60%

$$= \frac{(₹60\,crore)(0.096 - 0.093)(3/12)}{[1 + 0.096(3/12)]}$$

$$= \frac{(₹60\,crore)(0.00075)}{1.024} = ₹\ 4,39,453.13$$

Thus banker will pay Parker & Co. a sum of ₹ 4,39,453.13 or ₹ 4,39,453.

If actual rate of interest after 6 months happens to be 8.80%

$$= \frac{(₹60\,crore)(0.088 - 0.093)(3/12)}{[1 + 0.088(3/12)]}$$

$$= \frac{(₹60\,crore)(-0.00125)}{1.022} = -₹\ 7,33,855.19$$

Thus Parker & Co. will pay banker a sum of ₹ 7,33,855.19 or ₹ 7,33,855.

Note: Alternatively, students may solve the question on basis of different days count conventions instead of months (as considered in above calculations) i.e. 90 days/360 days, 90 days/ 365 days, 91 days/360 days or 91 days/365 days.

Practical Q2

2. TM Fincorp has bought a 6 x 9 ₹ 100 crore Forward Rate Agreement (FRA) at 5.25%. On fixing date reference rate i.e. MIBOR turns out be as follows:

Period	Rate (%)
3 months	5.50
6 months	5.70
9 months	5.85

You are required to determine:

(a) Profit/Loss to TM Fincorp. in terms of basis points.

(b) The settlement amount.

(Assume 360 days in a year)

Solution — ICAI

2. **(a)** TM will make a profit of 25 basis points since a 6X9 FRA is a contract on 3-month interest rate in 6 months, which turns out to be 5.50% (higher than FRA price).

(b) The settlement amount shall be calculated by using the following formula:

$$\frac{N(RR-FR)(dtm/360)}{1+RR(dtm/360)}$$

Where

N = Notional Principal Amount

RR = Reference Rate

FR = Agreed upon Forward Rate

Dtm = FRA period specified in days.

Accordingly:

$$\frac{100\,crore(5.50\%-5.25\%)(90/360)}{1+0.055(90/360)} = ₹\ 6,16,523$$

Hence there is profit of ₹ 6,16,523 to TM Fincorp.

Practical Q3

3. XYZ Limited borrows £ 15 Million of six months SONIA + 10.00% for a period of 24 months. The company anticipates a rise in SONIA, hence it proposes to buy a Cap Option from its Bankers at the strike rate of 8.00%. The lump sum premium is 1.00% for the entire reset periods and the fixed rate of interest is 7.00% per annum. The actual position of SONIA during the forthcoming reset period is as under:

Reset Period	SONIA
1	9.00%
2	9.50%
3	10.00%

You are required to show how far interest rate risk is hedged through Cap Option.

For calculation, work out figures at each stage up to three decimal points and amount nearest to £. It should be part of working notes.

Solution — ICAI

3. First of all we shall calculate premium payable to bank as follows:

$$P = \frac{rp}{\left[(1\div i) - \frac{1}{i\times(1+i)^t}\right]} \text{ X A or } \frac{rp}{PVAF(3.5\%,4)} \times A$$

Where

P = Premium

A = Principal Amount

rp = Rate of Premium

i = Fixed Rate of Interest

t = Time

$$= \frac{0.01}{\left[(1/0.035) - \frac{1}{0.035 \times 1.035^4}\right]} \times £15,000,000 \text{ or } \frac{0.01}{(0.966 + 0.933 + 0.901 + 0.871)}$$

$$\times £15,000,000$$

$$= \frac{0.01}{\left[(28.571) - \frac{1}{0.0402}\right]} \times £15,000,000 = £\ 40,595 \text{ or } \frac{£150,000}{3.671} = £\ 40,861$$

Now we see the net payment received from bank

Reset Period	Additional interest due to rise in interest rate	Amount received from bank	Premium paid to bank*	Net Amt. received from bank
1	£ 75,000	£ 75,000	£ 40,861	£34,139
2	£ 112,500	£ 112,500	£ 40,861	£71,639
3	£ 150,000	£ 150,000	£ 40,861	£109,139
TOTAL	£ 337,500	£ 337,500	£122,583	£ 214,917

Thus, from above it can be seen that interest rate risk amount of £ 337,500 reduced by £ 214,917 by using of Cap option.

* Alternatively, if premium paid is considered as £ 40,595, then above figure of £ 214,917 shall be changed to £ 215,715.

Practical Q4

4. Suppose a dealer quotes 'All-in-cost' for a generic swap at 8% against six month SOFR flat. If the notional principal amount of swap is US$ 5,00,000.

   (i) Calculate semi-annual fixed payment.

   (ii) Find the first floating rate payment for (i) above if the six month period from the effective date of swap to the settlement date comprises 181 days and that the corresponding SOFR was 6% on the effective date of swap.

   In (ii) above, if the settlement is on 'Net' basis, how much the fixed rate payer would pay to the floating rate payer?

   Generic swap is based on 30/360 days basis.

Solution — ICAI

4. (i) Semi-annual fixed payment

= (N) (AIC) (Period)

Where N = Notional Principal amount = US$ 5,00,000

AIC = All-in-cost = 8% = 0.08

$$= 5,00,000 \times 0.08 \left(\frac{180}{360}\right)$$

= 5,00,000 × 0.08 (0.5)

= 5,00,000 × 0.04 = ₹20,000/-

(ii) Floating Rate Payment

$$= N\,(SOFR) \left(\frac{dt}{360}\right)$$

$$= 5,00,000 \times 0.06 \times \frac{181}{360}$$

= 5,00,000 × 0.06 (0.503) or 5,00,000 × 0.06 (0.502777)

= 5,00,000 × 0.03018 or 5,00,000 × 0.030166 = US$ 15,090 or 15,083

(iii) Net Amount

= (i) – (ii)

= US$ 20,000 – US$ 15,090 = US$ 4,910

or = US$ 20,000 – US$ 15,083 = US$ 4,917

Practical QS

5. Derivative Bank entered into a plain vanilla swap through on OIS (Overnight Index Swap) on a principal of ₹ 10 crores and agreed to receive MIBOR overnight floating rate for a fixed payment on the principal. The swap was entered into on Monday, 2nd August, 2010 and was to commence on 3rd August, 2010 and run for a period of 7 days.

Respective MIBOR rates for Tuesday to Monday were:

7.75%,8.15%,8.12%,7.95%,7.98%,8.15%.

If Derivative Bank received ₹ 317 net on settlement, calculate Fixed rate and interest under both legs.

Notes:

(i) Sunday is Holiday.

(ii) Work in rounded rupees and avoid decimal working.

Solution — ICAI

5.

Day	Principal (₹)	MIBOR (%)	Interest (₹)
Tuesday	10,00,00,000	7.75	21,233
Wednesday	10,00,21,233	8.15	22,334
Thursday	10,00,43,567	8.12	22,256
Friday	10,00,65,823	7.95	21,795
Saturday & Sunday (*)	10,00,87,618	7.98	43,764
Monday	10,01,31,382	8.15	22,358
Total Interest @ Floating			1,53,740
Less: Net Received			317
Expected Interest @ fixed			1,53,423
Thus Fixed Rate of Interest			0.07999914
Approx.			8%

(*) i.e. interest for two days.

Note: Alternatively, answer can also be calculated on the basis of 360 days in a year.

Practical Q6

6. A Inc. and B Inc. intend to borrow $200,000 and $200,000 in ¥ respectively for a time horizon of one year. The prevalent interest rates are as follows:

Company	¥ Loan	$ Loan
A Inc	5%	9%
B Inc	8%	10%

The prevalent exchange rate is $1 = ¥120.

They entered in a currency swap under which it is agreed that B Inc will pay A Inc @ 1% over the ¥ Loan interest rate which the later will have to pay as a result of the agreed currency swap whereas A Inc will reimburse interest to B Inc only to the extent of 9%. Keeping the exchange rate invariant, quantify the opportunity gain or loss component of the ultimate outcome, resulting from the designed currency swap.

Solution — ICAI

6.

Opportunity gain of A Inc under currency swap	Receipt	Payment	Net
Interest to be remitted to B. Inc in $ 2,00,000 x 9% = $18,000 Converted into ($18,000 x ¥120)		¥21,60,000	
Interest to be received from B. Inc in $ converted into ¥ (6%x $2,00,000 x ¥120)	¥14,40,000	-	
Interest payable on ¥ loan	-	¥12,00,000	
	¥14,40,000	¥33,60,000	
Net Payment	¥19,20,000	-	
	¥33,60,000	¥33,60,000	
$ equivalent paid ¥19,20,000 x (1/¥120)			$16,000
Interest payable without swap in $			$18,000
Opportunity gain in $			$ 2,000

Opportunity gain of B inc under currency swap	Receipt	Payment	Net
Interest to be remitted to A. Inc in ($ 2,00,000 x 6%)		$12,000	
Interest to be received from A. Inc in ¥ converted into $ = ¥21,60,000/¥120	$18,000		
Interest payable on $ loan@10%	-	$20,000	
	$18,000	$32,000	
Net Payment	$14,000$32,000	- $32,000	
¥ equivalent paid $14,000 X ¥120			¥16,80,000
Interest payable without swap in ¥ ($2,00,000 X ¥ 120 X 8%)			¥19,20,000
Opportunity gain in ¥			¥ 2,40,000

Alternative Solution

Cash Flows of A Inc

(i) At the time of exchange of principal amount

Transactions		Cash Flows
Borrowings	$2,00,000 x ¥120	+ ¥240,00,000
Swap		- ¥240,00,000
Swap		+$2,00,000
Net Amount		+$2,00,000

(ii) At the time of exchange of interest amount

Transactions		Cash Flows
Interest to the lender	¥240,00,000X5%	¥12,00,000
Interest Receipt from B Inc.	¥2,00,000X120X6%	¥14,40,000
Net Saving (in $)	¥2,40,000/¥120	$2,000
Interest to B Inc.	$2,00,000X9%	-$18,000
Net Interest Cost		-$16,000

A Inc. used $2,00,000 at the net cost of borrowing of $16,000 i.e. 8%. If it had not opted for swap agreement the borrowing cost would have been 9%. Thus there is saving of 1%.

Cash Flows of B Inc

(i) At the time of exchange of principal amount

Transactions		Cash Flows
Borrowings		+ $2,00,000
Swap		- $2,00,000
Swap	$2,00,000X¥120	+¥240,00,000
Net Amount		+¥240,00,000

(ii) At the time of exchange of interest amount

Transactions		Cash Flows
Interest to the lender	$2,00,000X10%	- $20,000
Interest Receipt from A Inc.		+$18,000
Net Saving (in ¥)	-$2,000X¥120	- ¥2,40,000
Interest to A Inc.	$2,00,000X6%X¥120	- ¥14,40,000
Net Interest Cost		- ¥16,80,000

B Inc. used ¥240,00,000 at the net cost of borrowing of ¥16,80,000 i.e. 7%. If it had not opted for swap agreement the borrowing cost would have been 8%. Thus there is saving of 1%.

Practical Q7

7. A textile manufacturer has taken floating interest rate loan of ₹ 40,00,000 on 1st April, 2012. The rate of interest at the inception of loan is 8.5% p.a. interest is to be paid every year on 31st March. In the month of October 2012, the Central bank of the country releases following projections about the interest rates likely to prevail in future.

(i) On 31st March, 2013, at 8.75%; on 31st March, 2014 at 10% on 31st March, 2015 at 10.5% and on 31st March, 2016 at 7.75%. Show how the borrower can hedge the risk using Option Cap arising out of expected rise in the rate of interest when he wants to peg his interest cost at 8.50% p.a.

(ii) Assume that the premium negotiated by both the parties is 0.75% to be paid at once on 1st October, 2012 and the actual rate of interest on the respective due dates happens to be as: on 31st March, 2013 at 10.2%; on 31st March, 2014 at 11.5%; on 31st March, 2015 at 9.25%; on 31st March, 2016 at 8.25%. Show how the settlement will be executed on the perspective interest due dates.

Solution — ICAI

7. As borrower does not want to pay more than 8.5% p.a., on this loan where the rate of interest is likely to rise beyond this, hence, he has hedge the risk by entering into an agreement to buy interest rate caps with the following parameters:

- Notional Principal: ₹ 40,00,000/-
- Strike rate: 8.5% p.a.
- Reference rate: the rate of interest applicable to this loan
- Calculation and settlement date: 31st March every year
- Duration of the caps: till 31st March 2016
- Premium for caps: negotiable between both the parties

To purchase the caps this borrower is required to pay the premium upfront at the time of buying caps. The payment of such premium will entitle him with right to receive the compensation from the seller of the caps as soon as the rate of interest on this loan rises above 8.5%. The compensation will be at the rate of the difference between the rate of none of the cases the cost of this loan will rise above 8.5% calculated on ₹ 40,00,000/-. This implies that in none of the cases the cost of this loan will rise above 8.5%. This hedging benefit is received at the respective interest due dates at the cost of premium to be paid only once.

The premium to be paid on 1st October 2012 is **30,000/-** (₹ 40,00,000 x 0.75/100). The payment of this premium will entitle the buyer of the caps to receive the compensation from the seller of the caps whereas the buyer will not have obligation. The compensation received by the buyer of caps will be as follows:

On 31st March 2013

The buyer of the caps will receive the compensation at the rate of 1.70% (10.20 - 8.50) to be calculated on ₹ 40,00,000, the amount of compensation will be ₹ 68000/- (40,00,000 x 1.70/100).

On 31st March 2014

The buyer of the caps will receive the compensation at the rate of 3.00% (11.50 – 8.50) to be calculated on ₹ 40,00,000/-, the amount of compensation will be ₹ 120000/- (40,00,000 x 3.00/100).

On 31st March 2015

The buyer of the caps will receive the compensation at the rate of 0.75% (9.25 – 8.50) to be calculated on ₹ 40,00,000/-, the amount of compensation will be ₹ 30,000 (40,00,000 x 0.75/100).

On 31st March 2016

The buyer of the caps will not receive the compensation as the actual rate of interest is 8.25% whereas strike rate of caps is 8.5%. Hence, his interest liability shall not exceed 8.50%.

Thus, by paying once the premium upfront buyer of the caps gets the compensation on the respective interest due dates without any obligations.

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